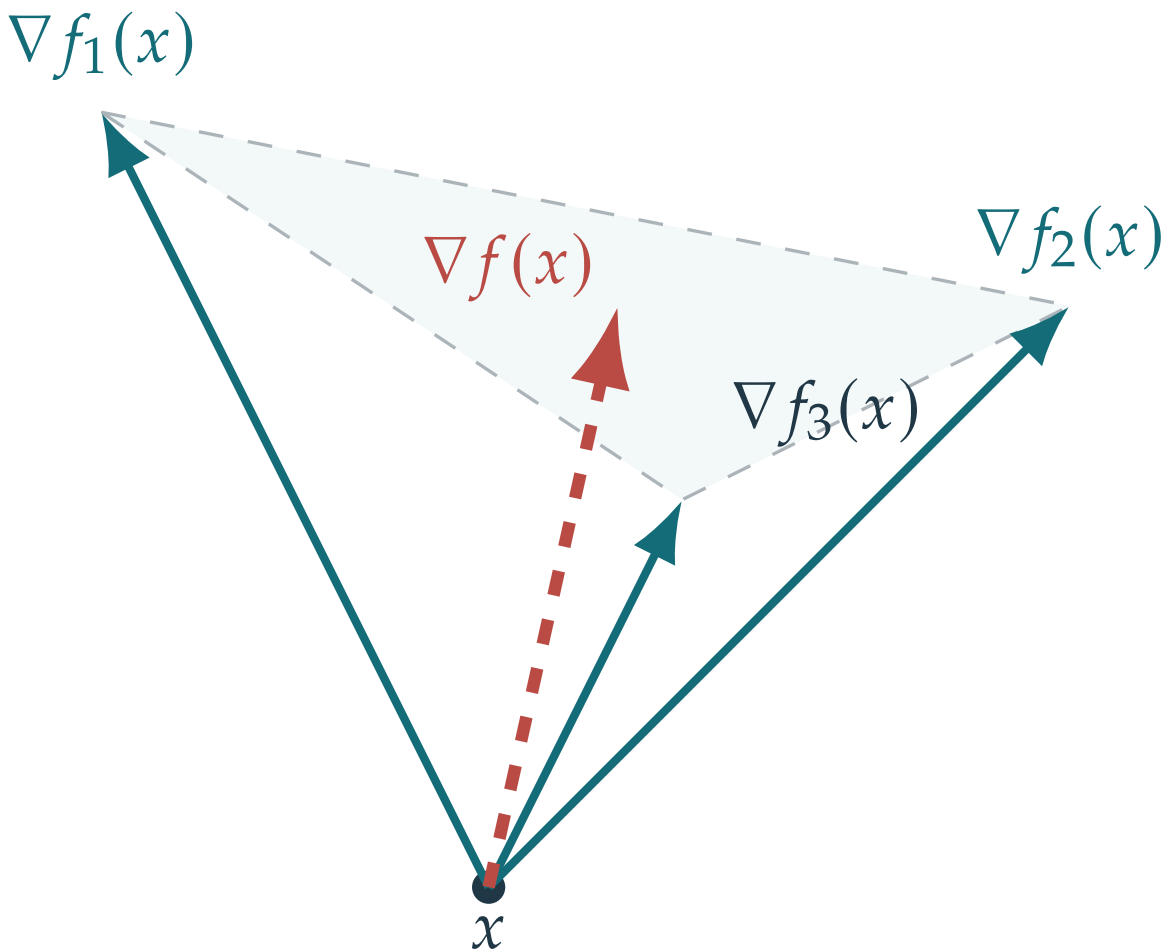
$$\mathbb{E}_i[\nabla f_i(x)] = \nabla f(x)$$



$$f = \tfrac{1}{3}(f_1 + f_2 + f_3)$$

$$i \sim \text{Unif}\{1, 2, 3\}$$